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Original article

Investigating the cooling of solar photovoltaic modules under the conditions of Riyadh

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ABSTRACT

Cooling enhances the energy conversion efficiency and output of photovoltaic (PV) panels. In this work, the effects of natural convection, forced convection, and evaporative cooling on the performance of polycrystalline PV panels were investigated. The output and efficiency of a cooled PV panel were monitored and compared to those of an uncooled PV panel under the same conditions. The cooling was conducted using an insulated channel installed below a PV panel. Natural convection cooling was investigated for various channel air gaps (H = 30, 60, 90, and 120 mm). Natural convection currents in the cooling channels were capable of cooling the panel with wide air gaps. In forced convection cooling, the air was introduced by fans installed at the bottom opening of the cooling channel with various air velocities (u_a = 1, 2, and 3 m/s). Evaporative natural convection cooling was performed by a wetted fabric along the lower surface of the cooling channel, whereas evaporative forced convection cooling by pushing air along the wetted lower surface of the channel. The experimental data showed that the panel efficiency and output increased due to cooling. The experimental results of natural convection cooling revealed that the use of an air gap of 120 mm to cool the solar panel contributed to an increase in the panel daily energy production and efficiency by 1.7% and 1.2%, respectively. For forced convection cooling, using air at a speed of 3 m/s increased the daily energy production by 4.4% and the efficiency by 4%. Natural convection evaporative cooling increased the daily energy production and the efficiency by 3.6% and 2.7%, respectively. Forced convection evaporative cooling contributed, at a speed of 2 m/s, to an increase in the daily energy production by 3.8% and an increase in efficiency of 3.8%.
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1. Introduction

The power received from the Sun is about 1.8 × 10^{11} MW, which equals 10,000 times the current energy demand rate of Earth (Parida et al., 2011; Mirzaei et al., 2014). Photovoltaic (PV) panels are used to convert solar energy into electrical energy. An ideal PV panel converts 6–20% of the incident solar irradiance into electricity (Dubey et al., 2013). However, nearly 50% of the incident solar energy is absorbed by the panels as heat (Chandrasekar et al., 2013). The absorbed thermal energy can raise the PV module temperature up to 40 °C above the surrounding temperature (Stropnik and Stritih, 2016; Makki et al., 2015). The increase in the temperature of the PV modules degrades panel efficiency and output power (Schwingshackl et al., 2013). Based on the PV cell type, the reduction in output power varies between 0.2 and 0.6% per degree rise in the module temperature (Home Energy, 2019). In addition, a high panel temperature for extended periods can cause delamination between its layers, glass breakage, bus bar (BB) failure, ethylene vinyl acetate (EVA) browning, moisture ingress, and diode failure (Bouaichi et al., 2019; Phinikarides et al., 2014; Rajput et al., 2016; Wohlgemuth et al., 2006) and can lead to a loss of 9.5 years from the module lifetime (Ogbomo et al., 2018). Finding a solution for the overheating of the PV module will give the confidence to the PV constructors and will accelerate the transformation process from the conventional to renewable photovoltaic energy. This transformation is expected to need 40–45 years (Zekry, 2020). Thus, the negative impact of the temperature

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Nomenclature

E	Daily energy (Wh)
E_{CP}	Generated energy from the cooled panel (Wh)
E_f	Energy consumed by the cooling fans (Wh)
E_{NP}	Generated energy from the uncooled panel (Wh)
H	Air gap height (mm)
I	Current (A)
k	Thermal conductivity (W/m·K)
P	Power (W)
T_{CP}	Average temperature of the cooled panel (°C)
T_{NP}	Average temperature of the uncooled panel (°C)
u_a	Air velocity (m/s)
V	Voltage (V)
W	Cooling channel width (m)

Subscripts

a	Air
av	Average
CP	Panel with isolated box (Cooled panel)
ex	Exit
f	Electrical fans
in	Inlet
max	Maximum
mp	Maximum power pint
NP	Normal panel (Non-colded panel)
oc	Open circuit
sc	Short circuit
w	Water

Greek symbols

ω	Humidity ratios (kg of water vapor/Kg of dry air)
ΔE	Daily energy difference between two panels (Wh)
ΔT_{av}	Two panels average temperature difference (°C)
ΔT_{max}	Maximum temperature difference between two panels (°C)
Δη	Efficiency difference (%)
μc-Si	Microcrystalline silicon
ṁ	Mass flow rate (kg/s)
β	Temperature coefficient of maximal power (%/°C)
η	Module efficiency (%)
ρ	Density (kg/m3)

Abbreviations

AM	Air mass
a-si	Amorphous silicon
BB	Electrical bus bar
BIPV	Building-integrated photovoltaic
CdTe	Cadmium telluride
CFD	Computational fluid dynamics
EVA	Ethylene vinyl acetate
KACST	King Abdulaziz City for Science and Technology
MC4	Multicontact connectors
m-si	Monocrystalline silicon
NOCT	Normal operating cell temperature (°C)
P-si	Polycrystalline silicon
PV	Photovoltaic
RTD	Resistance temperature detector
STC	Standard test conditions
XPS	Extruded polystyrene foam

increase on the module should be minimized to enhance PV module performance. For cold regions, PV modules can be cooled using natural convection. However, cooling PV modules installed in hot regions is challenging.

PV modules are very sensitive to relatively high ambient temperatures. The rise in the surrounding temperature reduces the output power and efficiency of PV modules. (Mazón-Hernández et al., 2013) investigated PV cooling by natural and forced convection using a rectangle steel channel installed beneath the PV module. They characterized the electrical behavior of the solar panels to improve their performance for roof installations. For natural convection, they found that the air gap of the cooling channel beneath the PV module must be wide enough to accommodate natural convection currents needed to cool the PV module. They concluded that natural and forced convection cooling increased the module peak power by 7.5% and 15%, respectively.

(Cuce et al., 2011) experimentally investigated the effects of passive cooling on the performance of silicon solar cells. They used an aluminum heat sink to dissipate heat from a PV cell. The dimensions of the heat sink were determined based on the steady-state heat transfer analysis. Their results showed that the performance of PV cells enhanced as the ambient temperature fell with and without the heat sink. At a solar radiation of 800 W/m^2, using the heat sink achieved an increase of 20% in the PV cell power output. The maximum cooling effect was observed at a solar radiation of 600 W/m^2.

The air gap beneath a PV module plays a primary role in removing heat from the panel via natural convection currents. Many numerical and experimental studies were conducted to determine the optimum air gap beneath a PV module to ensure optimal working conditions. (Gan, 2009) studied the air gap effect on the performance of the rooftop-installed PV modules using computational fluid dynamics (CFD) and found that the average module temperature decreased with increasing tilt angle and air gap. In addition, CFD results indicated that the average panel temperature decreased with increasing the module length to at least 0.08 m for air gaps. However, the maximum PV temperature increased with increasing panel length. To reduce the possible overheating of the PV modules, (Gan, 2009) suggested the minimum air gaps for single- and multi-panel installations to be 0.12–0.15 m and 0.14–0.16 m, respectively.

Building-integrated photovoltaics (BIPVs) installed on building roofs or walls play a key role in building shading and consequently, decrease air conditioning load. The air gap between the PV panel and roof or wall controls the convection heat transfer from the module to the roof and wall. The studies showed a 38% saving in the annual cooling load and a 2.5 °C decrease in the inner roof temperature under the PV module (Dominguez et al., 2011).

For BIPV systems, (Lau et al., 2012) investigated the effect of the air gap and inclination angle on PV module cooling. The buoyancy-driven flow in the cooling gaps below the PV modules was considered similar to the flow in a heated open-ended channel. In this investigation, the channel inclination angle ranged from 15° to 90°, and a channel height-to-gap ratio of 20 was used. (Lau et al., 2012) found that the open-ended channels with small inclination angles were characterized by a small chimney effect, which damped the natural currents. Therefore, the heat transfer along the PV panels decreased for small inclination angles.

(Moshfegh and Sandberg, 1998) investigated natural convection along the lower surface of PV panels. In their numerical and experimental investigations, they considered heat exchange by radiation using a module with a length of 7.0 m and an air gap of 0.23 m. Dif-

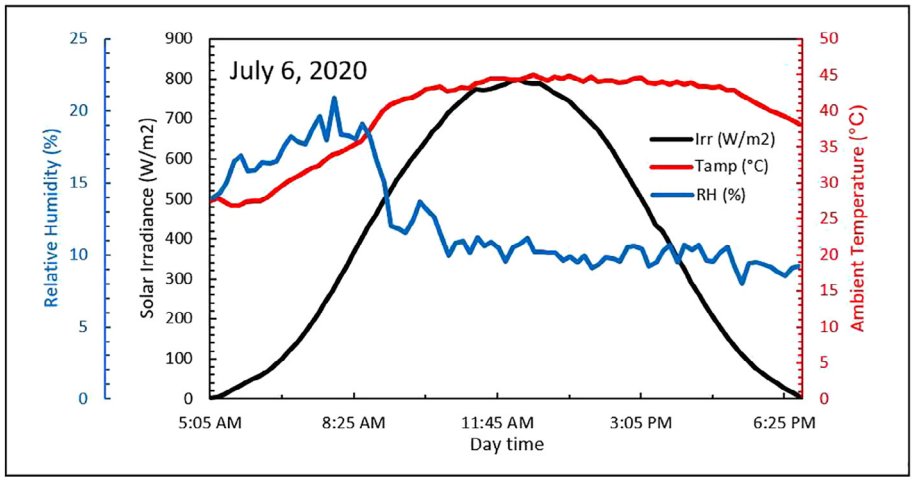


Fig. 1. Ambient temperature, relative humidity, and the global radiation on July 6, 2020, in Riyadh, Saudi Arabia.

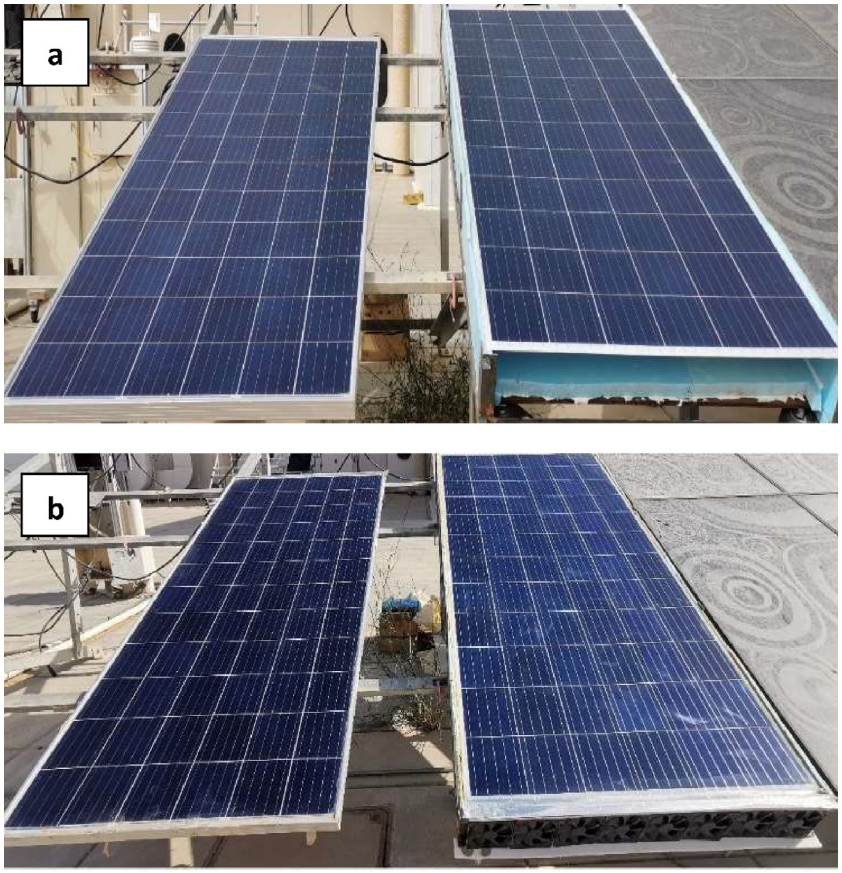


Fig. 2. PV module test setup. (a) Natural convection and natural convection with evaporation experiments, (b) forced convection and forced convection with evaporation experiments. The panels on the left are uncooled, and those on the right are cooled using the cooling channel.

ferent input heat fluxes and emissivity of the bounding surfaces were considered. They concluded that the surface under the PV panel received 30% of the heat emitted from the solar panel while the rest was dissipated through convection.

(Schwingshackl et al., 2013) studied the effect of wind on PV module cooling. For several cell types, they investigated the effect of the wind speed and direction and ambient temperature on cell temperature and output power. They found that the more wind was present; the power plant became more efficient. Also, they calculated the maximum normal operating cell temperature (NOCT) for different module technologies. They found that polycrystalline and amorphous silicon modules had the maximum normal operating cell temperature (NOCT). Moreover, they considered the power degradation due to temperature. They found that the polycrystalline module had the highest power degradation due to the module temperature increase. For this type of panel, the data of (Schwingshackl et al., 2013) indicated a power degradation of 0.45% per degree rise in the module temperature.

The performances of PV panels in a windy location were compared with those in an unwindy location by (Gökmen et al.,

Table 1
Mechanical Characteristics of Panels.

Solar Cells	Polycrystalline silicon 156.75 × 156.75 mm (6 in.), 5 Bus Bar
No. of Cells	72 (6 × 12)
Dimensions	1,956 mm × 992 mm × 40 mm (77 × 39.05 × 1.57 in.)
Weight	22 kg (48 lbs.)
Front Glass	3.2 mm tempered glass
Frame	Anodized aluminum alloy
Junction Box	IP67 rated (3 bypass diodes)
Cables	4.0 mm2 (0.006 in. 2), symmetrical length 1,000 mm (39.4 in.)
Connectors	MC4 connectors

Table 2
Electrical Characteristics of Panels.

STC (Standard test conditions; 1000 W/m^2, 25 °C, AM1.5)	Model # KS-320PC
Optimum Operating Voltage (V_{mp}) (V)	36.39
Optimum Operating Current (I_{mp}) (A)	8.8
Open Circuit Voltage (V_{oc}) (V)	44.95
Short Circuit Current (I_{sc}) (A)	9.24
Maximum Power @ STC (P_{max}) (W)	320
Module Efficiency (%)	16.5
Operating Module Temperature (°C)	− 40 to + 85
Maximum System Voltage (VDC)	1,000
Maximum Series Fuse Rating (A)	20
Power Tolerance	±3%
Maximum Power at NOCT (P_{max})	237
Optimum Power Voltage (V_{mp})	34.64
Optimum Operating Current (I_{mp}) (A)	6.84
Open Circuit Voltage (V_{oc}) (V)	43.04
Short Circuit Current (I_{sc})	7.29
Nominal Operating Cell Temperature (NOCT) (°C)	45 ± 2
Temperature Coefficient of P_{max} (%/°C)	−0.400
Temperature Coefficient of V_{oc} (%/°C)	−0.330
Temperature Coefficient of I_{sc} (%/°C)	0.058

2016). They also studied the effect of changing the tilt angle of the panel for various durations from 15 min up to a year. Their data indicated that the maximum relative energy difference increased by about 3.55% for tilt angle moving periods of 15 min, 30 min, and one hour.

(Teo et al., 2012) conducted experiments to examine PV panel performance with and without active cooling. They observed that without active cooling, the temperature of the modules was high, and a solar cell efficiency in the range 8–9% was achieved. However, when the module was cooled by forced convection, the temperature significantly dropped, leading to an increase in the efficiency between 12% and 14%. (Teo et al., 2012) developed a model to predict the module temperature, and the comparison between the measured and predicted temperature profiles indicated good agreement.

For natural convection evaporative PV panel cooling, (Chandrasekar et al., 2013) developed a simple passive evaporative cooling system with a cotton wick installed on the bottom surface of the PV panel. They studied the effect of evaporative cooling on the thermal and electrical performances of the PV module. The cooling system consisted of a cotton wick wetted by water, Al_2O_3/water nanofluid, or CuO/water nanofluid. The temperature of the PV module was reduced to 59 °C and 54 °C. These results corresponded to a reduction of 11% and 17%, respectively, in module temperature compared to the module temperature obtained without cooling.

(Alami, 2014) investigated a passive evaporative cooling technique to control the temperature rise of the PV modules due to the absorption of solar irradiance using a synthetic clay element supported at the back of the module that allowed a thin film of water to be evaporated. The results indicated that the output voltage of the PV modules increased by 19.4% and the power output by 19.1%.

(Haidar et al., 2018) experimentally studied the effect of the evaporative natural convection cooling of a PV panel under Riyadh's outdoor conditions. The evaporative cooling was achieved by attaching a piece of fabric to the bottom surface of the PV panel. This fabric was kept wet by dripping water using rubber hoses attached to the panel's bottom surface. The latent evaporation heat was utilized to absorb heat from the body of the tested PV module. They achieved a 20 °C reduction in PV panel temperature and a 14% increase in electrical power generation efficiency compared with a similar PV module operated under the same conditions.

Cooling using air especially with forced convection is a good solution. For example (Salaudeen, 2019) evaluated the performance of a thermal energy storage system with the use of paraffin wax as the storage medium, and air as the heat transfer fluid. Salaudeen (2019) found that the air was the ideal heat transfer fluid due to its availability and durability.

Forced convection with evaporation cooling was investigated by (Haidar et al., 2016). They developed a theoretical model for concurrent air and water films in a cooling channel installed below the PV panel. The model simulated heat and mass transfer that occurred at the bottom of the PV panel. The results revealed the effects of geometrical parameters, air flow rate, temperature, and humidity on the cooling process. They concluded that when the air inlet temperature decreased, the temperature of the PV panel slightly decreased. A reduction of 6 °C in the PV panel temperature was obtained with cooling.

In this study, the performances of PV modules using various cooling techniques, namely natural convection, forced convection, natural convection with evaporation, and forced convection with evaporation were experimentally investigated. The effects of these cooling methods on the daily energy produced by the panel, peak efficiency of the panel, and panel temperature were investigated.

2. Experimental setup

The experiments were carried out in a testing facility (KACST PV Laboratory) located in Riyadh, Saudi Arabia, which has significant high temperatures and dry weather conditions. The tilt angle of the PV panels was set at 23°, which is the optimal fixed tilt angle for Riyadh City. The experiments were conducted from March to September which covers the hot dry conditions of Riyadh City. Fig. 1 shows a sample of the climate data collected by the weather plant of King Abdulaziz City for Science and Technology. This weather station (VSN800 Weather Station with advanced sensor set) located in KACST PV Lab was used to obtain needed weather conditions. It was installed 2 m away from the experimental setup. Fig. 1 shows the ambient temperature, relative humidity, and horizontal global radiation for typical daytime solar radiation with a peak value around noon. The relative humidity and dry-bulb temperature reflect the typical Riyadh weather in the summer.

Two identical PV panels with 72 cells and 5 BBs were used. The uncooled PV panel had an aluminum metal frame, and the cooled panel had no metal frame. The first panel was used to examine PV performance without cooling. The second panel was supported above an isolated channel, as shown in Fig. 2, to examine how cooling affects its performance. The two panels were identical and had typical power outputs. The panels' data given by the manufacturer are listed in Tables 1 and 2.

Initial power measurements were conducted to ensure that the two panels were properly working and had typical power output values. The verification was carried out using a Sun simulator under standard test conditions (STC: Irradiation = 1000 W/m^2, panel temperature = 25 °C AM = 1.5). The results are shown in Fig. 3. The measured output powers of the panels were 309.2 and

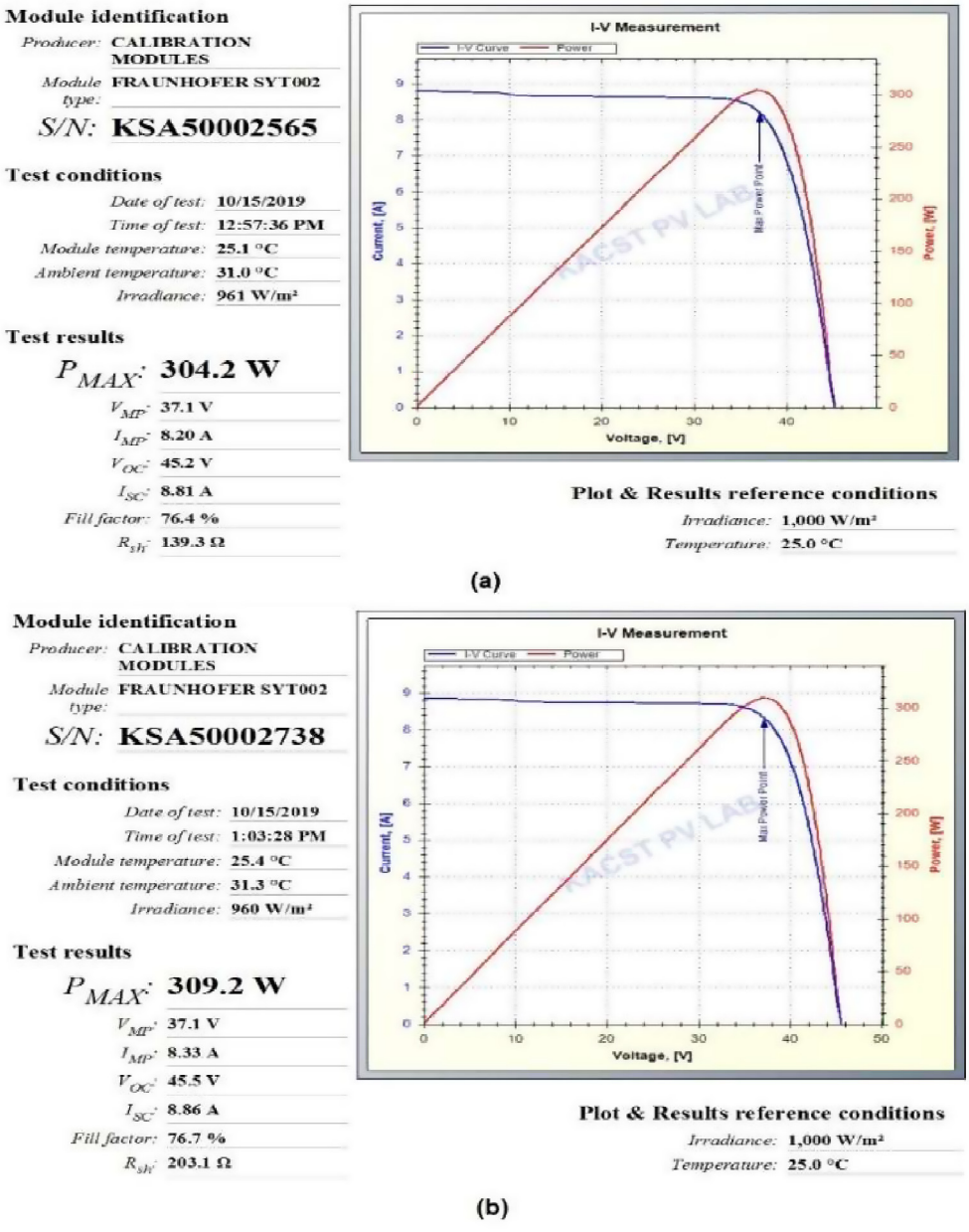


Fig. 3. I–V curve under STC for (a) uncooled panel (b) cooled panel.

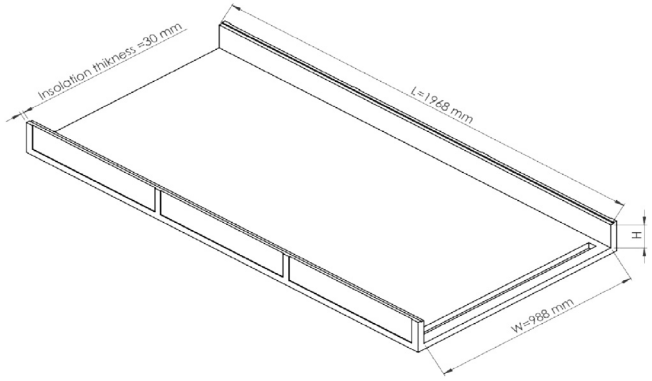


Fig. 4. Panel cooling channel dimensions.

304.2 W. This deviation is acceptable since the manufacturer allows a 3% deviation in power measurement at STC.

The dimensions of the cooling channel installed below the second panel are shown in Fig. 4. The open-ended cooling channel was made of an extruded polystyrene foam (XPS) board with an adjustable height (H). A thermal insulation board (Thermal conductivity = 0.029 W/m·K (An, et al., 2017)) was used to build this channel to minimize heat transfer through channel walls. In natural convection and natural convection with evaporation experiments, the bottom and top ends of the cooling channel were open, as shown in Fig. 2a. For forced convection and forced convection with evaporation experiments, the bottom end was equipped with fans, as shown in Fig. 2b.

Calibrated T-type thermocouples were used to measure panels' back surface temperature. Nine thermocouples were attached to the back of each panel, as shown in Fig. 5. In addition, six thermocouples were used to measure the ambient temperature, air tem-

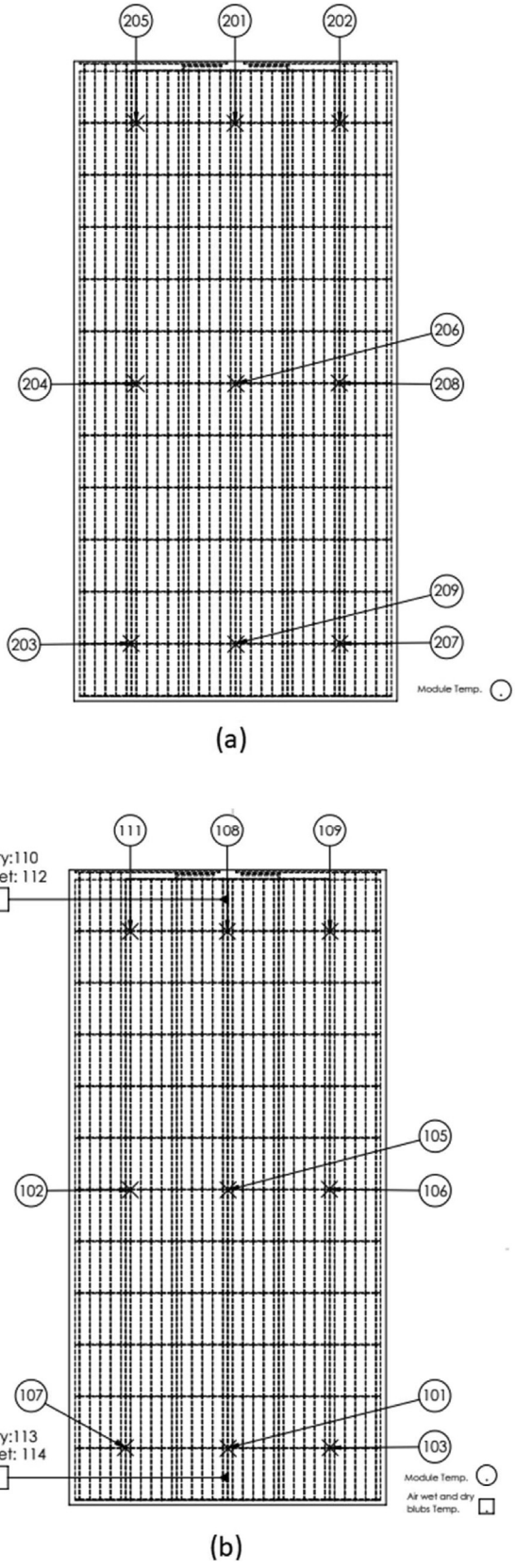


Fig. 5. Thermocouple locations and corresponding channel number for a) uncooled panel and b) cooled panel.



Fig. 6. Calibration setup for the thermocouples.

Table 3
Experimental instruments and uncertainties.

Instrument or Parameter	Unit	Uncertainty
T- type Thermocouples	°C	±0.125%
Air Speed Indicator (Pacer DA400 Anemometer)	m/s	±12.5%
Panel Efficiency	%	±1.3%
PV panel Energy	Wh	±0.1%
Water Consumption	L/h	±12.6%

peratures along the channels, and wet-bulb temperatures. Fig. 5 shows the location and given numbers of the thermocouples.

The thermocouples used were calibrated using a temperature calibration unit (ISOTECH Hyperion 936 Basic) equipped with a high accuracy RTD200 temperature sensor reference with an uncertainty of ±0.05%. The calibration data indicated an uncertainty of ±1.2 °C in the readings of the thermocouples.

Temperature data were collected using a data logger (Agilent 34972A), which recorded the measurement every 2 min. The temperature data were recorded from sunrise to sunset.

The air velocity indicator (Pacer DA400 anemometer) was used to measure air velocity at the cooling channel inlet and exit.

Microinverter (Enphase IQ7) was used to generate loads for the PV panels. It converted DC to AC and delivered it to the grid system. This module was equipped with a maximum power point tracking (MPPT) system that allows the extraction of the maximum power from the PV panels. Enphase IQ7 microinverter had a built-in monitoring system, which indicates the actual produced power and energy, which was used to monitor daily energy produced by the panels.

A thermal camera (FLIR-T440) was used to produce the thermal images of the PV panel surfaces at the peak of the Sun irradiation (around 1:00 pm), i.e., at the time when the system was stable and at the highest temperature.

Axial flow fans were used to force the air through the channel. They were used in forced convection and forced convection with evaporative experiments.

For the experiment with evaporation, a fabric was spread along the bottom surface of the cooling channel. This fabric was kept wet during the experiment. A submerged water pump was used to circulate the water from the bottom water tank to the upper edge of the cooling channel exit, where 12 nozzles were used to spread water on the fabric.

To evaluate the accuracy of the main experimental findings, the uncertainty propagation function built in EES software, Eq. (1), was

Table 4
Summary of natural convection experiments.

Date	H (mm)	E (Wh)		$\frac{\Delta E}{E_{NP}}$ (%)	η (%)		$\frac{\Delta\eta}{\eta_{Np}}$(%)	ΔT_{max} (°C)	ΔT_{av} (°C)	u_a (m/s)	Mean			
		E_{CP}	E_{NP}		η_{CP}	η_{NP}					$\frac{\Delta E}{E_{NP}}$(%)	$\frac{\Delta\eta}{\eta_{Np}}$(%)	ΔT_{max} (°C)	ΔT_{av} (°C)
8-Mar-20	30	1848	1842	0.3	13.7	13.9	−1.5	−3.3	−1.2	0.3–0.9	0.6	−0.7	−1.6	−0.8
9-Mar-20		1971	1947	1.2	13.6	13.5	0.4	−1.9	0.2					
10-Mar-20		1951	1933	0.9	13.4	13.4	−0.4	−0.7	0.2					
8-Jun-20		1692	1695	−0.2	14.6	14.8	−1.3	−0.7	−2.4					
6-Mar-20	60	1861	1852	0.5	13.6	13.6	−0.4	0.5	−0.1	0.3–0.4	1.2	−0.2	−1.4	0.0
7-Mar-20		1624	1608	1.0	16.4	16.7	−1.5	−3.6	−0.8					
11-Mar-20		1954	1913	2.1	13.2	13.1	0.8	1.0	1.6					
7-Jun-20		1732	1715	1.0	14.7	14.6	0.4	−3.5	−0.8					
3-Mar-20	90	2095	2053	2.0	13.7	13.5	1.1	2.2	2.1	0.5–1.6	1.5	0.6	0.1	1.3
4-Mar-20		1978	1957	1.1	13.3	13.3	0.4	0.5	1.3					
5-Mar-20		1941	1916	1.3	13.4	13.4	0.0	−1.7	1.3					
6-Jun-20		1778	1750	1.6	15.0	14.8	0.9	−0.5	0.3					
12-Mar-20	120	1816	1771	2.5	13.0	12.8	1.5	3.1	2.1	0.7–1.2	1.7	1.2	2.0	1.2
13-Mar-20		1638	1613	1.5	14.8	14.6	1.2	3.0	1.5					
14-Mar-20		785	775	1.3	15.1	14.8	2.0	1.8	0.9					
4-Jun-20		1776	1754	1.3	14.8	14.8	0.0	−0.1	0.1					

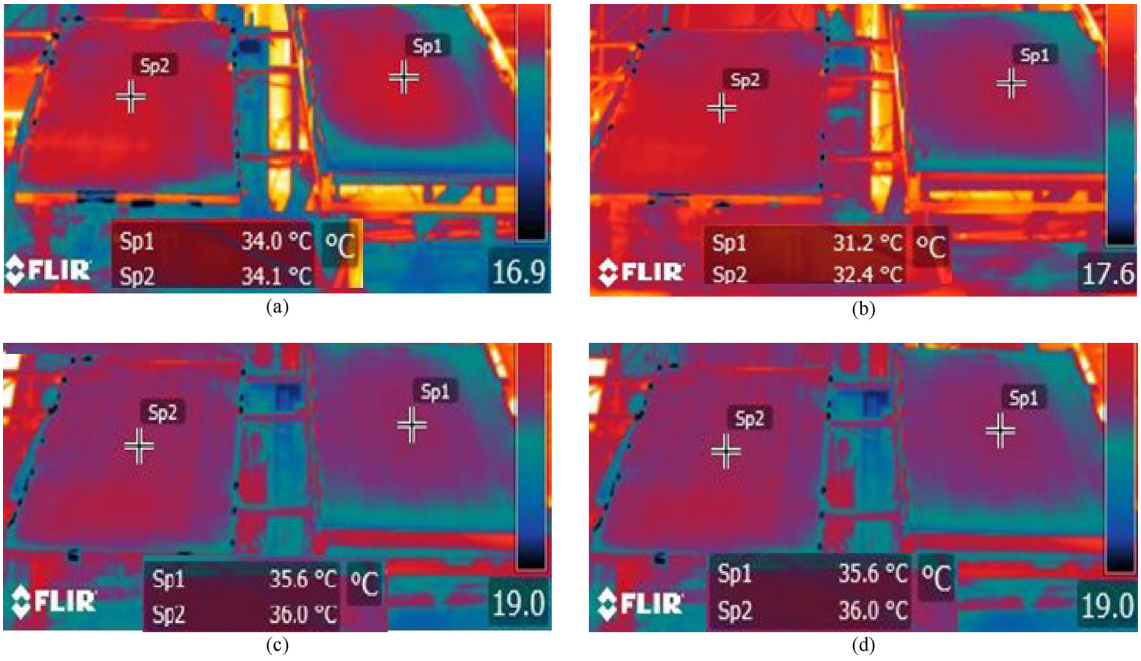


Fig. 7. Thermal images of natural convection cooled and uncooled panels for a) H = 30 mm, b) H = 60 mm, c) H = 90 mm, and d) H = 120 mm.

used. The used sensors were calibrated before performing the experiments to estimate their uncertainties.

Assuming the individual measurements are uncorrelated and random, the uncertainty in the calculated quantity can be determined from,

$$U_Y = \sqrt{\sum_i \left(\frac{\partial Y}{\partial X_i}\right)^2 U_{X_i}^2} \quad (1)$$

where U represents the uncertainty of the variable Y.

The thermocouples used in the experiments were calibrated using a temperature calibration unit (ISOTECH Hyperion 936 Basic) equipped with a high accuracy RTD200 temperature sensor reference as shown in Fig. 6. Based on the calibration data and for the temperature range from 20 to 80 °C, the uncertainty in temperature measurement was ±0.125%.

Considering the above procedure for experimental parameters, the uncertainties are listed in Table 3.

3. Results and discussion

3.1. Natural convection cooling

The effect of natural convection cooling on PV panel performance was investigated for different cooling channel gaps (H = 30, 60, 90, and 120 mm). A summary of the experimental data is listed in Table 4. This table shows the data collected for the cooled and uncooled panels. For each air gap, the experiment was collected for a period of 4–7 days to get minimum ambient air velocity as much as possible. From the collected data, the data of the day with the lowest wind speed were considered in the data analysis.

The thermal images of the two panels' upper surfaces are shown in Fig. 7 for different air gaps. The thermal images indicated that the natural convection currents (chimney effect) in the cooling channel were capable of cooling the panel better than that without the cooling channel. Each PV panel had nine thermocouples attached to its back surface. The temperature of the panel was esti-

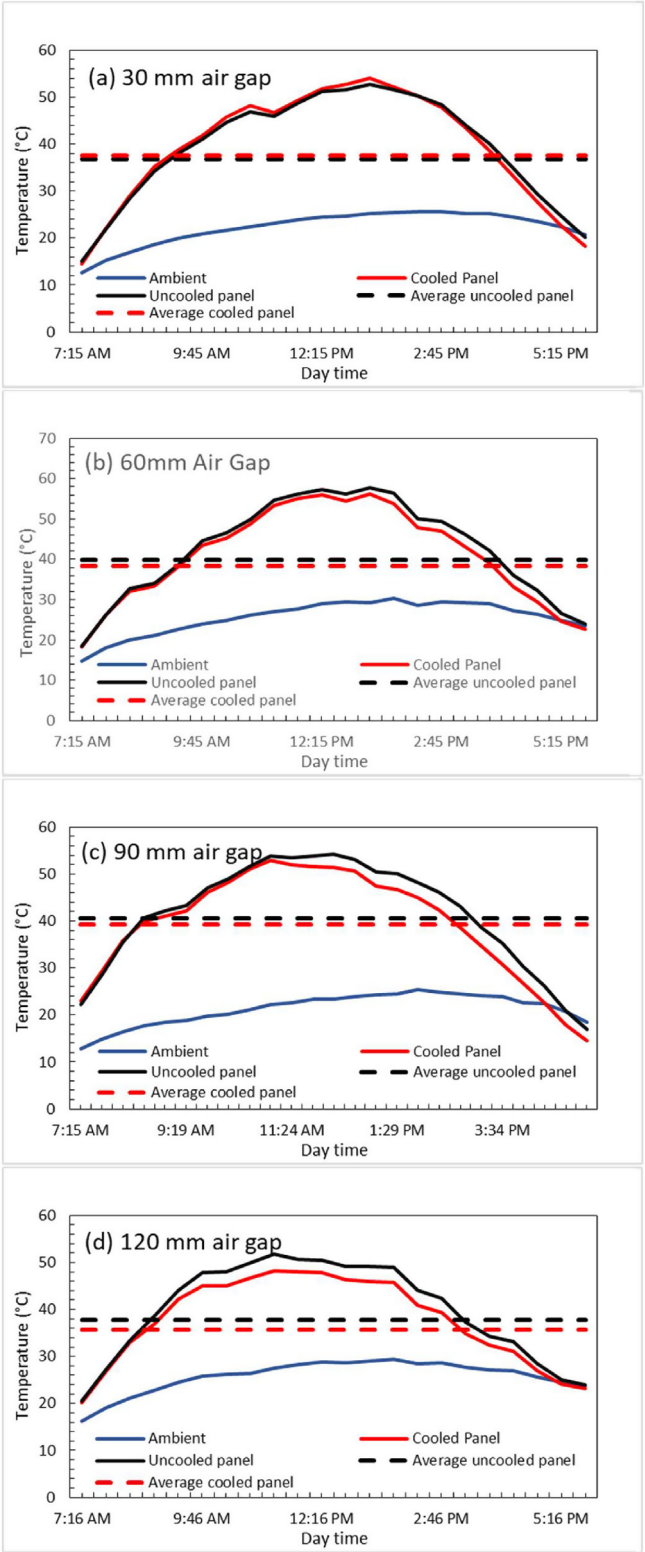


Fig. 8. Temperature histories of natural convection cooled and uncooled panels for a) H = 30 mm, b) H = 60 mm, c) H = 90 mm, and d) H = 120 mm.

mated by averaging the readings of these nine thermocouples. The data collected for each air gap in four days are listed in Table 4. Fig. 8 shows panel temperatures, the blue curve represents the ambient temperature, the red curve represents the cooled panel temperature, and the black curve represents the uncooled panel temperature. The horizontal dashed lines display the average panel temperatures for the cooled panel (red) and uncooled panel (black). As shown in these figures, the panel temperatures reached the maximum values between 11:00 am and 2:00 pm, which was about 30 °C higher than the ambient temperature. As shown in Table 4 and Fig. 8, the daily average temperature difference (ΔT_{av}) between the cooled and uncooled panels ranged from −0.8 to 1.2 °C. The highest daily average temperature drop was observed for a 120-mm air gap channel with a temperature drop of 1.2 °C.

The drop in the maximum temperature (ΔT_{max}) of the cooled panel compared to the uncooled panel was calculated by

$$\Delta T_{max} = T_{NPmax} - T_{CPmax} \quad (2)$$

where T_{NPmax} is the maximum temperature of the uncooled panel, and T_{CPmax} is the maximum temperature of the cooled panel.

As shown in Table 4, Figs. 7 and 8, the temperature of the cooled panel was slightly lower than the uncooled panel, except for the ones with gaps of 30 and 60 mm. This temperature difference increased as the air gap increased. However, for the narrow air gaps (H = 30 and 60 mm), the friction in the cooling channel resisted natural convection currents. Consequently, the natural convection around the uncooled panel was more effective than that inside the cooling channel. As a result, utilizing channels with 30- and 60-mm air gaps reduced cooling efficiency. The data in Table 4 show a variation ranging from −1.6 to 2 °C in the maximum temperature difference. The highest temperature difference (2 °C) occurred for the 120-mm air gap.

The efficiency of the panels calculated during the power peak time was calculated from

$$\eta = \frac{P/A}{Irr} \times 100 \quad (3)$$

where η is the efficiency of the panel (%), P is the power (W), A is the area of the panel (m^2), and Irr is the solar irradiance (W/m^2).

The efficiency improvement due to the cooling was estimated from

$$\frac{\Delta\eta}{\eta_{Np}}\% = \frac{\eta_{CP} - \eta_{Np}}{\eta_{Np}} \times 100 \quad (4)$$

where $\Delta\eta$ is the efficiency difference (%), η_{CP} is the efficiency of the cooled panel (%), and η_{Np} is the efficiency of the uncooled panel (%).

It was found that the panel efficiency increased as the air gap increased. The maximum efficiency occurred at an air gap of 120 mm with an improvement of 1.2%. For H = 30 and 60 mm, the efficiency of the cooled panel was lower than that of the uncooled panel since the cooling process was unable to cool the panel.

The microinverters (Enphase IQ7) were used to generate loads for the PV panels and monitor the daily generated energy. Table 4 shows the difference in the daily produced energy (ΔE) between the cooled and uncooled panels. The energy enhancement percentage caused by the cooling process was calculated from

$$\frac{\Delta E}{E_{NP}}\% = \frac{E_{CP} - E_{NP}}{E_{NP}} \times 100 \quad (5)$$

where $\Delta E/E_{NP}\%$ is the energy enhancement percentage, E_{CP} is the daily energy generated from the cooled panel, and E_{NP} is the daily energy generated from the uncooled panel.

As shown in Table 4, the highest energy gain percentage was obtained for H = 120 mm, where the cooled panel produced 1.7% more energy compared to the uncooled panel. On the other hand, the lowest was 0.6%, which occurred for H = 30 mm.

Table 5
Summary of 12 days of forced convection experiments (H = 120 mm).

Date	u_a (m/s)	E (Wh)		$\frac{\Delta E}{E_{NP}}$ (%)	η (%)		$\frac{\Delta\eta}{\eta_{Np}}$ (%)	ΔT_{max} (°C)	ΔT_{av} (°C)	Mean			
		E_{CP}	E_{NP}		η_{cp}	η_{Np}				$\frac{\Delta E}{E_{NP}}$(%)	$\frac{\Delta\eta}{\eta_{Np}}$(%)	ΔT_{max} (°C)	ΔT_{av} (°C)
20-Mar-20	1	1656	1614	2.6	14.3	14.0	1.9	5.0	2.0	1.9	0.6	2.2	1.2
27-Mar-20		1764	1740	1.4	16.9	17.0	−0.4	−0.3	0.2				
28-Mar-20		1504	1478	1.8	13.9	13.9	0.0	2.7	1.3				
1-Jun-20		1825	1790	2.0	15.3	15.2	0.8	1.2	1.1				
18-Mar-20	2	1622	1580	2.7	15.8	15.4	2.4	5.6	2.6	3.0	2.4	7.6	3.8
19-Mar-20		1967	1898	3.6	14.6	14.1	3.4	11.1	5.6				
29-Mar-20		1916	1858	3.1	14.7	14.4	2.7	8.3	4.7				
2-Jun-20		1817	1770	2.7	15.3	15.1	1.3	5.2	2.4				
16-Mar-20	3	2099	1982	5.9	15.4	14.4	6.3	16.2	9.1	4.4	4.0	11.3	6.0
17-Mar-20		1917	1857	3.2	16.0	15.6	2.6	6.7	4.3				
30-Mar-20		1990	1903	4.6	14.7	14.1	4.2	12.4	6.7				
3-Jun-20		1848	1782	3.7	15.2	14.8	2.9	10.0	3.8				

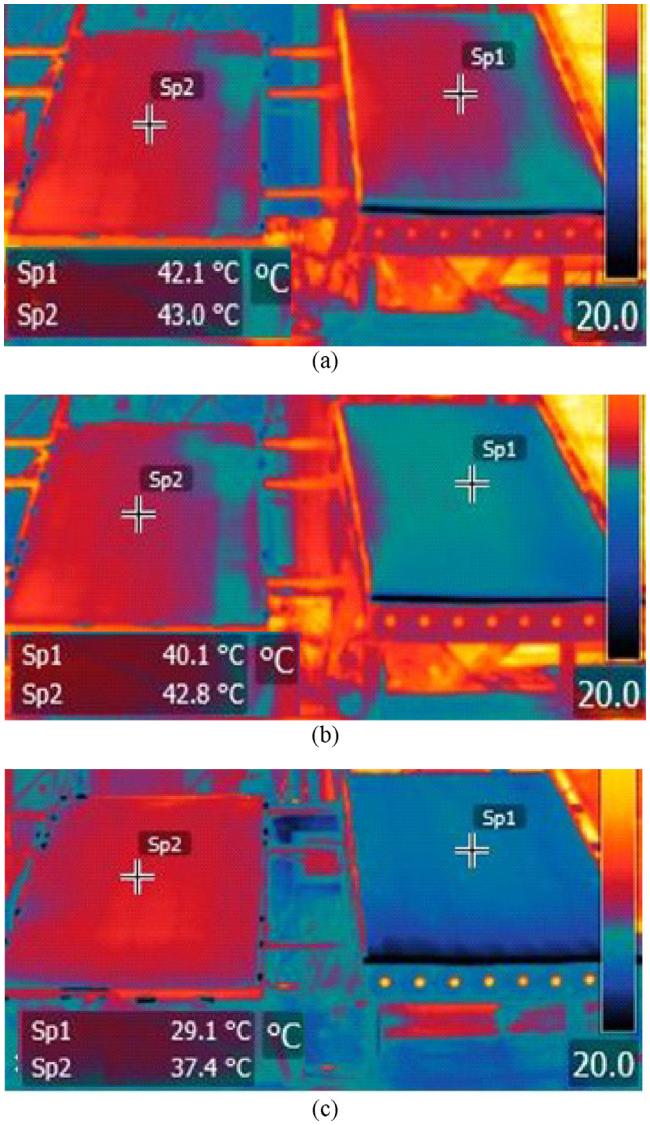


Fig. 9. Thermal images of panel surfaces for forced convection cooled and uncooled panels at air velocities of a) 1 m/s, b) 2 m/s, and c) 3 m/s.

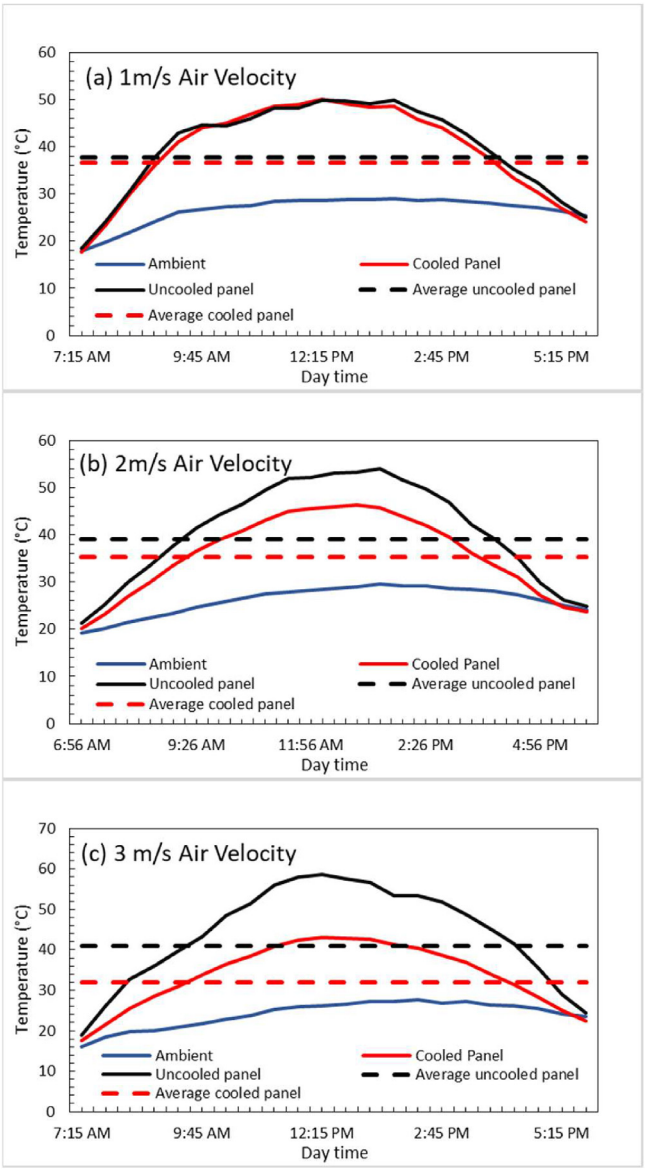


Fig. 10. Temperature histories of forced convection cooled and uncooled panels at air velocities of a) 1, b) 2, and c) 3 m/s.

3.2. Forced convection cooling

The effect of forced convection on the performance of the PV panels was studied. The cooling channel width was fixed at H = 120 mm during these experiments. The cooling channel bottom opening was equipped with eight axial flow fans to push the air from the bottom to the top of the channel. These fans were operated using a monitored DC power source to supply the necessary power and measure the consumed energy. The air fans con-

Table 6
Summary of natural convection with evaporation experiments.

Date	H (mm)	E (Wh)		$\frac{\Delta E}{E_{NP}}$ (%)	η (%)		$\frac{\Delta\eta}{\eta_{Np}}$(%)	ΔT_{max} (°C)	ΔT_{av} (°C)	$\dot{m}_w$ (l/h)	u_a (m/s)	Mean				
		E_{CP}	E_{NP}		η_{cp}	η_{Np}						$\frac{\Delta E}{E_{NP}}$ (%)	$\frac{\Delta\eta}{\eta_{Np}}$(%)	ΔT_{max} (°C)	ΔT_{av} (°C)	$\dot{m}_w$ (l/h)
30-Jun-20	30	1783	1742	2.4	15.1	14.9	1.3	3.0	3.2	0.8	0.38–0.42	2.5	1.6	3.2	3.5	0.8
1-Jul-20		1773	1734	2.2	15.0	14.8	1.3	2.5	3.3	1.0						
2-Jul-20		1771	1722	2.8	15.3	15.0	2.2	3.9	4.0	0.7						
3-Jul-20		1799	1756	2.4	15.1	14.9	1.7	3.5	3.4	0.6						
26-Jun-20	60	1842	1786	3.1	15.0	14.6	2.5	5.1	4.9	1.6	0.93–1.61	3.2	2.1	5.2	4.7	1.3
27-Jun-20		1859	1800	3.3	15.3	14.9	2.5	5.7	4.9	0.3						
28-Jun-20		1835	1781	3.0	15.2	14.9	1.7	4.8	4.3	1.7						
29-Jun-20		1836	1777	3.3	15.3	15.0	1.7	5.1	4.5	1.7						
22-Jun-20	90	1843	1780	3.5	15.5	15.2	2.1	4.3	4.0	0.8	0.62–1.39	3.4	2.5	5.0	4.2	0.8
23-Jun-20		1825	1768	3.2	15.5	15.1	2.9	5.3	4.1	0.8						
24-Jun-20		1854	1787	3.7	15.6	15.2	2.5	5.0	4.5	–						
25-Jun-20		1781	1728	3.1	15.7	15.3	2.6	5.6	4.3	–						
18-Jun-20	120	1868	1810	3.2	15.3	15.0	2.5	4.9	4.3	1.4	0.24–0.3	3.6	2.7	5.3	4.5	1.2
19-Jun-20		1881	1811	3.9	15.3	14.8	3.3	4.6	4.6	1.4						
20-Jun-20		1834	1765	3.9	15.7	15.2	3.0	5.7	4.9	1.0						
21-Jun-20		1696	1640	3.4	15.4	15.1	2.1	5.9	4.3	–						

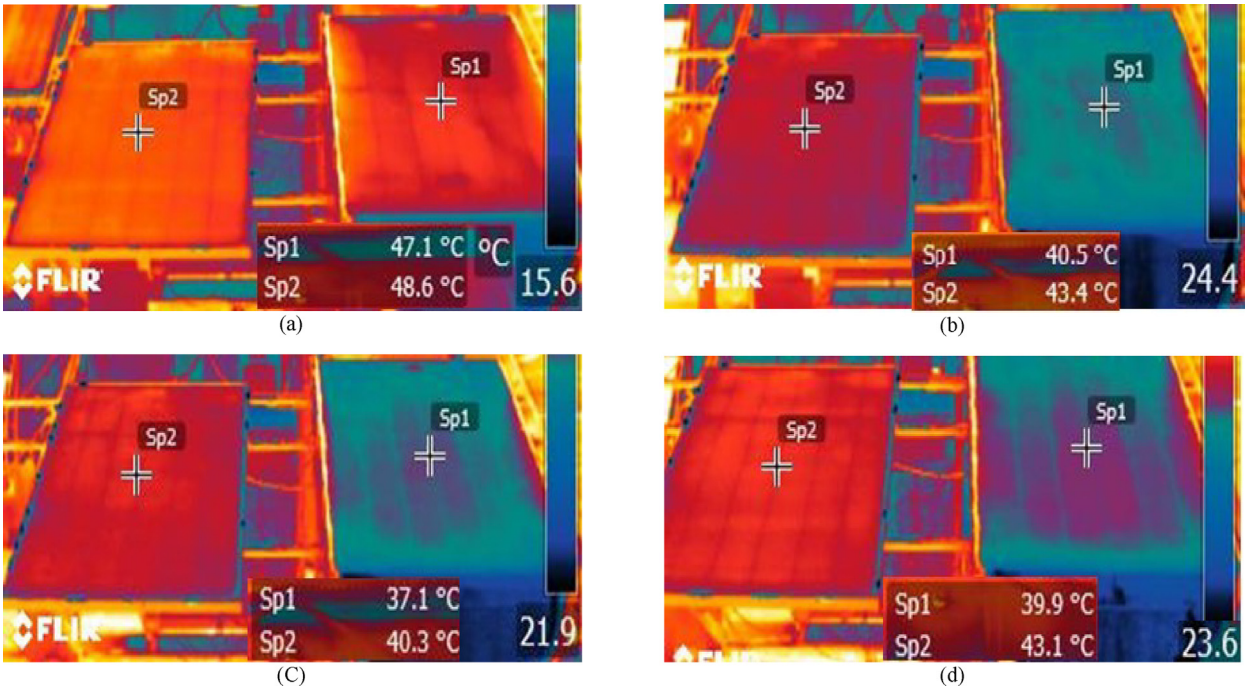


Fig. 11. Thermal images for evaporative natural convection cooled and uncooled panels: a) H = 30 mm, b) H = 60 mm, c) H = 90 mm, and d) H = 120 mm.

sumed approximately 0.56 kWh per day. This value was excluded from energy calculations. The performance of the PV panel was examined for three air velocities (1, 2, and 3 m/s). Table 5 contains a summary of 12 days forced convection experiments.

The thermal images for the PV panel surfaces are shown in Fig. 9. These images were captured at the peak of the power production at 1:00 pm. The air velocity in the cooling channel affects the panel temperature, as can be seen in Figs. 9 and 10. A significant drop in the temperature of the cooled panel can be observed on the right of the thermal images as the air velocity increased to 3 m/s and the panel temperature dropped to 11 °C.

As discussed before, the temperatures of panels were estimated based on the readings of the thermocouples attached to their back surface. The panel temperature histories for the examined air velocities are shown in Fig. 10. The forced convection cooling effect improved as the air velocity increased from 1 to 3 m/s. As indicated in Table 5 and Figs. 9 and 10, the temperature of the cooled panel was lower than the uncooled panel in the examined air velocities. The maximum temperature difference (ΔTmax) between the cooled and uncooled panel was about 2.2, 7.6, and 11.3 °C for air velocities of 1, 2, and 3 m/s, respectively. For the daily average temperature difference (ΔTav) between the cooled and uncooled panels, the cooling by forced convection reduced the temperature to 1.2, 3.8, and 6 °C at air velocities of 1, 2, and 3 m/s, respectively, as shown in Fig. 10.

The panel temperature drop was reflected in the efficiency improvement ($\frac{\Delta\eta}{\eta_{Np}}$) of the cooled panel compared to the uncooled panel. As shown in Table 5, the efficiency of the cooled panel

was higher than the uncooled panel by 0.6, 2.4, and 4.0% for air velocities of 1, 2, and 3 m/s, respectively.

Table 5 shows the difference between the cooled and uncooled panels in daily produced energy ($\frac{\Delta E}{E_{NP}}$). As indicated in Table 5, the panel cooled at an air velocity of 3 m/s has the highest energy efficiency compared to the uncooled panel, where a 4.4% enhancement in the daily energy output was achieved.

3.3. Natural convection cooling with evaporation

Natural convection with evaporation was performed for different air gaps by keeping the bottom surface of the cooling channel wet. For each air gap (H = 30, 60, 90, and 120 mm), the experiments were conducted for at least four days. The experiment procedure was similar to that performed in the natural convection experiments, except that the lower surface of the channel was wet. A layer of fabric sheet was placed on the lower surface of the channel to increase the evaporation effect. A water spray system was used to keep the fabric surface wet using a circulation water pump. The water pump was consuming nearly 0.14 kWh per day. This value was excluded from energy calculations. The wet-bulb temperatures at the channel inlet and exit were measured using the wet-bulb temperature sensors. The humidity ratio was found using EES software at measured dry- and wet-bulb temperatures. The water evaporation rate was estimated from

$$\dot{m}_w = \dot{m}_a(\omega_{ex} - \omega_{in}) \quad (6)$$

where $\dot{m}_w$ is the water evaporation flow rate (kg/s), ω_{in} and ω_{ex} are the humidity ratios at the cooling channel inlet and exit, respectively, and $\dot{m}_a$ is the air mass flow rate in the channel (kg /s). The air mass flow rate ($\dot{m}_a$) was estimated from

$$\dot{m}_a = \rho_a H W u_a \quad (7)$$

where ρ_a is the air density (kg/m3) at channel exit conditions, u_a is the air velocity (m/s) measured in the channel, H is the channel height (H = 0.03, 0.06, 0.09, or 0.12 m), and W is the channel width (1.96 m).

Table 6 shows a summary of evaporative natural convection experiments. The thermal images in Fig. 11 and the temperature histories of panels shown in Fig. 12 indicated the effectiveness of evaporation on PV cooling compared to natural convection without evaporation. The heat transfer from the cooled panel was enhanced because of the air humidification process. As shown in Table 6 and Fig. 12, the cooling using evaporative natural convection reduced the maximum temperature from 3.2 to 5.3 °C compared to the uncooled panel. For the daily average temperature difference (ΔT_{av}) between the cooled and uncooled panels, the cooling by natural convection with evaporation helped to increase ΔT_{av} to 3.5–4.7 °C. The difference in panel temperatures was reflected in the efficiency improvement ($\frac{\Delta\eta}{\eta_{Np}}$). As shown in Table 6, the efficiency of the cooled panel increased from about 1.6 to 3.6%. Table 6 indicates that the daily energy produced from the cooled panel was enhanced by 2.5–3.6% for air gaps of 30 and 120 mm, respectively.

3.4. Forced convection cooling with evaporation

Panel cooling using forced convection with evaporation effect was studied for a channel gap of 120 mm for air velocities of 1, 2, and 3 m/s. In this experiment, axial fans were used to push air in the cooling channel while its bottom surface was maintained wet, as discussed in the previous section.

Table 7 indicates a summary of the results of the forced convection with evaporation experiments. The consumed water ($\dot{m}_w$) during the experiments is presented in Table 7. The water consumption varied from 1.2 to 4.9 l/h.

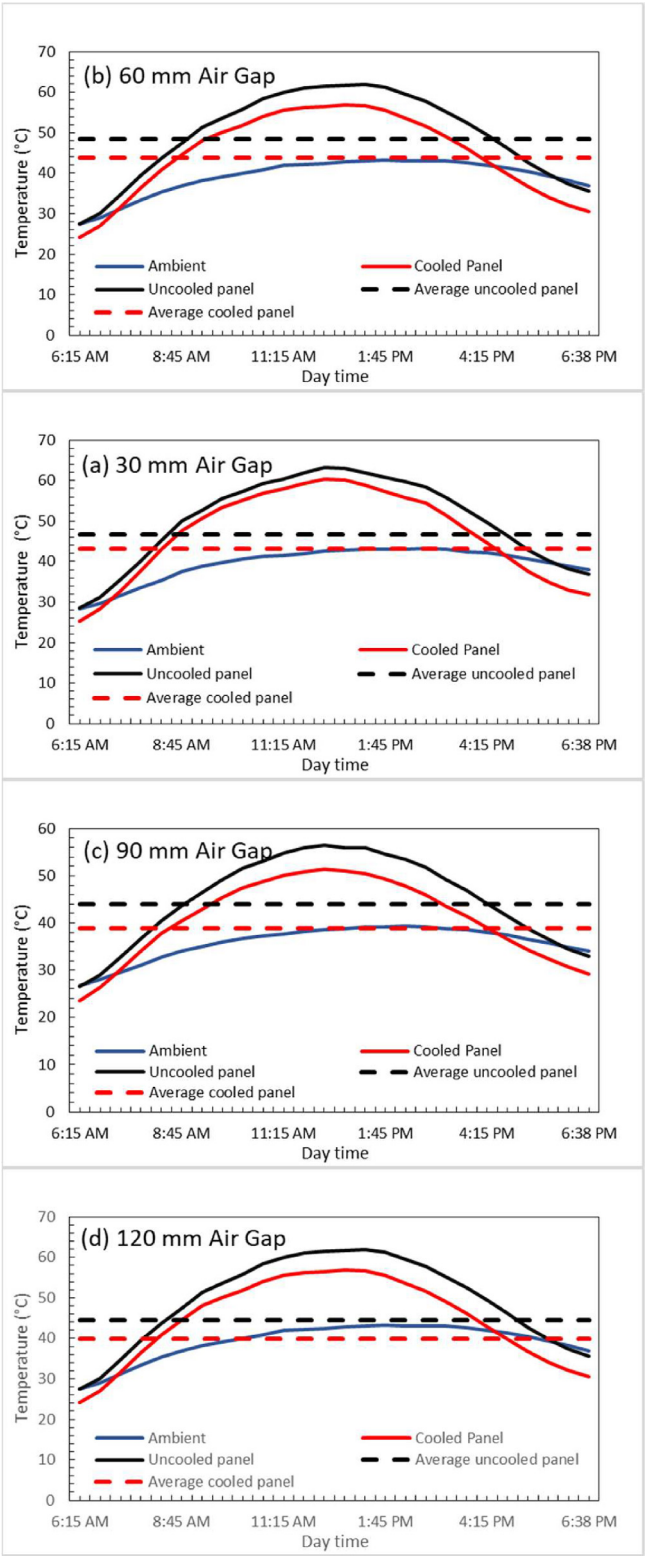


Fig. 12. Temperature histories of evaporative natural convection cooled and uncooled panels at air gaps of a) 30, b) 60, c) 90, and d) 120 mm.

The thermal images in Fig. 13 show the effect of air velocity on the temperatures of cooled and uncooled panels. Moreover, Fig. 14 shows the temperature history for different air velocities. A significant drop in the maximum temperature (ΔT_{max}) was observed from 7.3 to 12.5 °C for air velocities of 1 and 3 m/s, respectively. Compared to the data of forced convection without evaporation,

Table 7
Summary of forced convection with evaporation experiments (H = 120 mm).

Date	u_a (m/s)	E (Wh)		$\frac{\Delta E}{E_{NP}}$ (%)	η (%)		$\frac{\Delta\eta}{\eta_{Np}}$(%)	ΔT_{max} (°C)	ΔT_{av} (°C)	$\dot{m}_w$ (l/h)	Mean				
		E_{CP}	E_{NP}		η_{cp}	η_{Np}					$\frac{\Delta E}{E_{NP}}$ (%)	$\frac{\Delta\eta}{\eta_{Np}}$(%)	ΔT_{max} (°C)	ΔT_{av} (°C)	$\dot{m}_w$ (l/h)
5-Jul-20	1	1811	1747	3.7	15.4	15.0	2.6	9.1	5.1	0.8	3.1	2.5	7.3	4.6	1.2
6-Jul-20		1810	1733	4.4	15.3	14.7	3.9	9.3	6.1	0.6					
7-Jul-20		1798	1732	3.8	15.1	14.7	2.6	9.2	5.5	0.7					
8-Jul-20		1831	1758	4.2	15.2	14.7	3.0	7.9	5.1	0.4					
19-Sep-20		1831	1788	2.4	12.5	12.3	1.6	6.6	3.9	2.5					
20-Sep-20		1868	1828	2.2	12.4	12.2	2.0	5.6	3.8	1.6					
21-Sep-20		1731	1692	2.3	12.4	12.1	1.9	4.7	3.9	0.8					
22-Sep-20		1702	1667	2.1	12.4	12.1	2.0	5.8	3.5	2.4					
9-Jul-20	2	1847	1762	4.8	15.4	14.7	4.8	11.7	6.5	0.5	3.7	3.5	10.3	6.0	2.1
10-Jul-20		1810	1730	4.6	15.4	14.7	4.6	10.0	5.9	0.9					
11-Jul-20		1712	1640	4.4	15.5	14.9	4.0	9.1	5.4	1.3					
12-Jul-20		1761	1696	3.8	15.2	14.8	2.6	7.4	4.9	1.9					
12-Sep-20		1871	1823	2.6	12.3	12.0	2.4	10.2	5.4	2.3					
14-Sep-20		1864	1800	3.6	12.4	11.9	4.0	12.6	7.0	3.5					
15-Sep-20		1873	1822	2.8	12.5	12.1	2.8	9.4	6.5	3.8					
16-Sep-20		1870	1813	3.1	12.4	12.1	2.8	11.9	6.4	2.7					
14-Jul-20	3	1616	1555	3.9	14.9	14.4	3.3	12.3	6.2	3.0	3.8	3.8	12.5	6.4	4.9
15-Jul-20		1741	1667	4.4	15.2	14.7	3.5	12.3	6.2	3.9					
16-Jul-20		1797	1737	3.5	15.5	15.0	3.0	11.6	5.3	5.1					
17-Jul-20		1872	1786	4.8	15.2	14.6	4.2	12.6	6.8	4.9					
8-Sep-20		1908	1850	3.1	12.9	12.5	3.5	11.5	6.6	8.6					
9-Sep-20		1889	1817	4.0	12.4	11.9	4.0	14.1	7.3	7.7					
10-Sep-20		1647	1590	3.6	10.5	10.0	5.0	14.4	5.8	1.3					
11-Sep-20		1643	1588	3.5	12.3	11.8	3.8	11.5	6.6	4.4					

evaporation facilitates panel cooling. The average daily panel temperature drop (ΔT_{av}) observed was 4.6 °C at an air velocity of 1 m/s and increased to 6.4 °C at an air velocity of 3 m/s. The average temperature values are presented in Table 7 and Fig. 14.

The drop in panel temperatures was reflected on the efficiency improvement ($\frac{\Delta\eta}{\eta_{Np}}$) of the cooled panel. As shown in Table 7, the efficiency of the cooled panel was higher than that of the uncooled panel by 2.5, 3.5, and 3.8% for 1, 2, and 3 m/s, respectively. The summarized results in Table 7 show enhancement of 3.1, 3.7, and 3.8% in the daily produced energy at 1, 2, and 3 m/s, respectively.

4. Comparison of cooling techniques

The comparison of experimental data obtained on the investigated four cooling techniques revealed that the cooled panel temperature decreased as the air gap and velocity increased. In addition, the evaporative cooling assisted in reducing the panel temperature.

In Fig. 15, the maximum temperature drops of the cooled panel compared to the uncooled panel is presented. The data in the figure shows how the peak temperature of the cooled panel drops due to cooling. In forced convection cooling with evaporation (green bars), the cooled panel temperature decreased by 12 °C compared to the uncooled panel. The daily average panel temperature dropped with the proposed cooling methods, as shown in Fig. 16, except for those with natural convection with air gaps of 30 mm and 60 mm.

The critical parameter that should be addressed in these experiments was the efficiency of the PV panel. In Fig. 17, the improvement in panel efficiency is shown. The data in the figure indicate an improvement of up to 4% in panel efficiency using forced convection at an air velocity of 3 m/s.

The comparison of the daily average energy is shown in Fig. 18. In general, the examined cooling techniques assisted in increasing the produced energy. The enhancement in the daily energy varied from 0.5 to 4.5% for the cooled panel compared to the uncooled panel.

5. Conclusions

The performances of polycrystalline PV panels were investigated under natural convection, forced convection, and evaporative natural and evaporative forced convection cooling techniques. In natural convection cooling, the cooling was enhanced as the cooling channel width increased. For the 120-mm air gap cooling channel, a maximum daily energy enhancement of 1.7%, an improvement in the panel efficiency of 1.2%, a maximum temperature drop of 2 °C, and an average daily temperature drop of 1.2 °C were observed.

Forced convection cooling was improved as the air velocity increased. A maximum daily energy enhancement of 4.4%, a panel efficiency improvement of 4.0%, a maximum temperature drop of 11.3 °C, and an average daily temperature drop of 6 °C were observed at an air velocity of 3 m/s.

Evaporative natural convection enhanced the cooling process compared to natural convection cooling. An energy output enhancement of 3.6%, a panel efficiency improvement of 2.7%, a maximum temperature drop of 5.3 °C, and an average daily temperature drop of 4.5 °C were observed for an air gap of 120 mm.

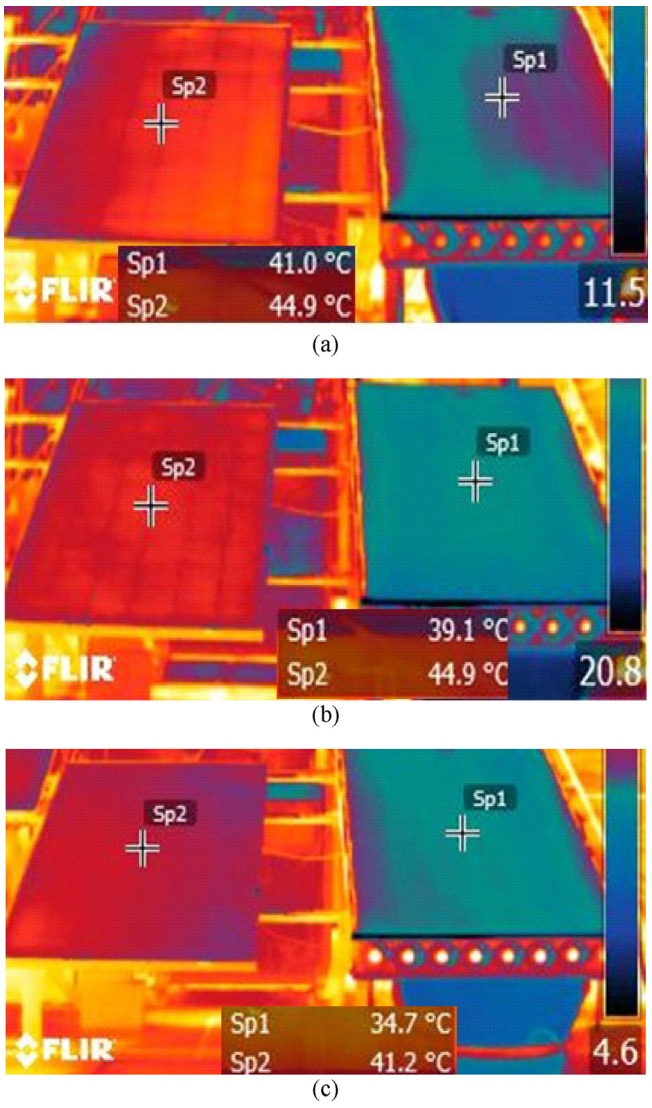


Fig. 13. Thermal images of evaporative forced convection cooled and uncooled panels at air velocities of a) 1 m/s, b) 2 m/s, c) 3 m/s.

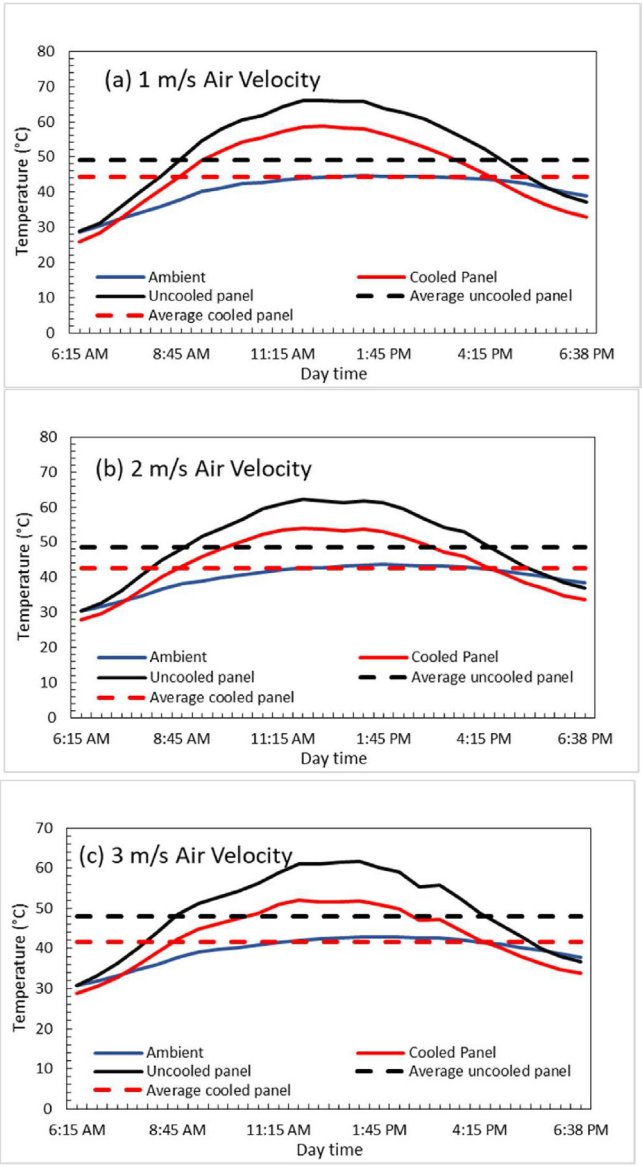


Fig. 14. Temperature histories of evaporative forced convection cooled and uncooled panels at air velocities of a) 1, b) 2, and c) 3 m/s.

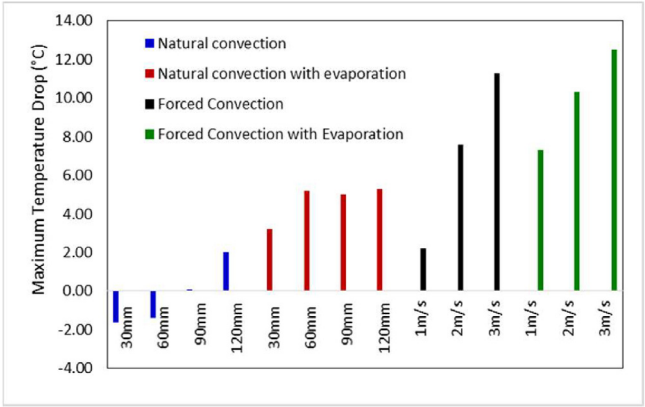


Fig. 15. Comparison of the maximum temperature drops for four cooling techniques.

Evaporative forced convection cooling improved panel cooling compared to forced convection cooling. At an air velocity of 3 m/s, a maximum temperature drop of 12.5 °C, a daily average panel temperature drop of 6.4 °C, an energy increase of 3.8%, and a panel efficiency improvement of 3.8% were observed.

Among all, the forced convection cooling at an air velocity of 3 m/s showed the highest improvement in daily energy production (4.4%) and panel efficiency (4.0%).

Further investigations may be needed for better PV module cooling. Further investigations may include enhancing evaporation technique, installing wetted fabrics on the back of the PV module and investigating effect of air velocity on water droplet carry over.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

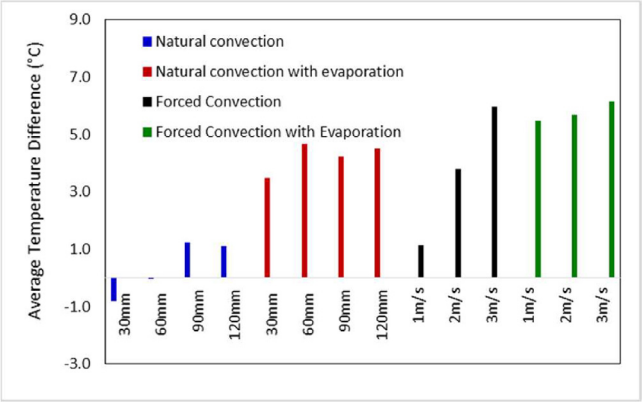


Fig. 16. Comparison of average temperature drops for four cooling techniques.

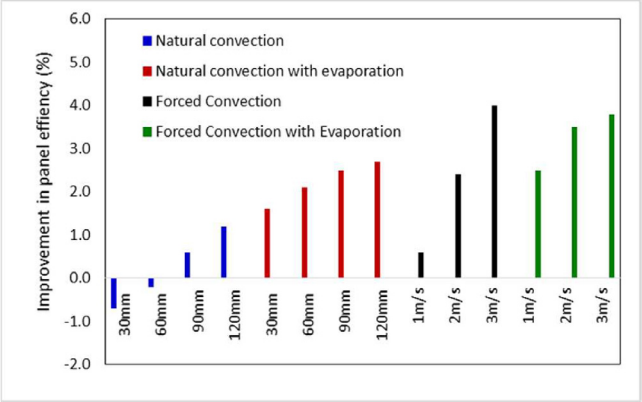


Fig. 17. Comparison of panel efficiency improvement for four cooling techniques.

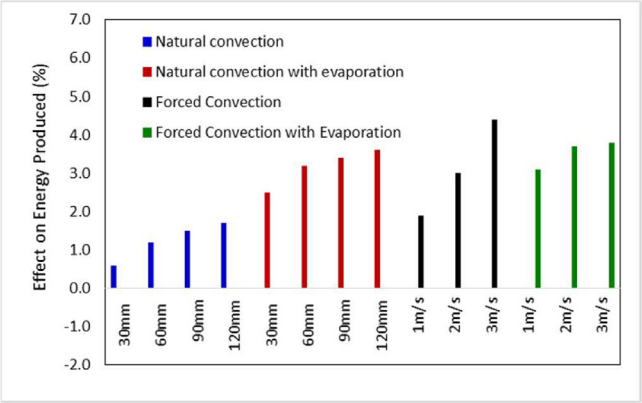


Fig. 18. Comparison of enhancement in produced energy for four cooling techniques.

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